

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0152

Clover 345 kV
June 25, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0152. The Interconnection Customer proposes a 334.5 MW natural gas combined cycle facility interconnecting at the Clover 345 kV substation in the Virginia area. The request was received on August 31, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 104 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,083,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Bauxite-Klamath Falls 138kV line
Contingency: Clover-Bear 345kV
Loading: 114.7% (Rating: 253 MVA)
DFAX: 7.6%
Pre-project loading: 104.7%
2. Facility: Klamath Falls-Kenbridge Substation 138kV line
Contingency: Bear-Clover Substation 345kV
Loading: 114.4% (Rating: 248 MVA)
DFAX: 7.6%
Pre-project loading: 104.2%
3. Facility: Kenbridge-Keokuk 138kV Line
Contingency: Bear-Clover 345kV
Loading: 112.9% (Rating: 247 MVA)
DFAX: 7.6%
Pre-project loading: 102.6%
4. Facility: Hutchins-Bauxite 138kV
Contingency: Bear Substation-Clover 345kV
Loading: 104.5% (Rating: 329 MVA)
DFAX: 10.0%
Pre-project loading: 94.3%
5. Facility: Littlefork-Lake Lure 138kV L
Contingency: Fluvanna P.S-Louisa Generation Facility 345kV
Loading: 104.4% (Rating: 494 MVA)
DFAX: 6.4%
Pre-project loading: 100.0%

6. Facility: Lake Lure-Bluff City 138kV
Contingency: Fluvanna P.S-Louisa Generation Facility 345kV
Loading: 102.3% (Rating: 408 MVA)
DFAX: 5.2%
Pre-project loading: 98.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Klamath Falls-Bauxite 138kV line	Rebuild line with higher-capacity conductor	\$6,946,000	100% (\$6,946,000)
Klamath Falls-Kenbridge 138kV	Rebuild line with higher-capacity conductor	\$8,756,000	100% (\$8,756,000)
Kokuk-Kenbridge 138kV	Rebuild line with higher-capacity conductor	\$7,800,000	100% (\$7,800,000)
Hutchins-Bauxite 138kV	Reconductor line	\$1,747,000	100% (\$1,747,000)
Littlefork-Lake Lure 138kV L	Reconductor line	\$2,922,000	10% (\$289,000)
Bluff City-Lake Lure 138kV	Reconductor line	\$5,786,000	9% (\$545,000)
	Total allocated to project		\$26,083,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.